

Milica Radivojević

Electrical & Computer Engineering Graduate | Computer Engineering & Control Systems

Novi Sad, Serbia • milica.radivojevic03@gmail.com • +381 642673596 • linkedin.com/in/mradivojevic03

EDUCATION

B.Sc. in Electrical and Computer Engineering (240 ECTS) - GPA: 8.97/10

Faculty of Technical Sciences, University of Novi Sad, 2022-2026. Graduated September 2026.

Specialization: Computer engineering and Control systems

EXPERIENCE

• Software Development Intern – RT-RK

April – September 2026.

- Developed a reusable NestJS server plugin for telemetry validation and normalization within a metadata-driven IoT system.
 - Built a dynamic React UI plugin enabling automated, metadata-driven dashboard generation.
 - Integrated front and backend plugins into host applications as part of a bachelor thesis.
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TECHNICAL SKILLS

- **Embedded:** AUTOSAR Classic, Linux, Bash, RTOS, SSH, RPi, Arduino, ADC, UART, SPI, I²C
 - **Programming/Software:** C/C++, Java, Kotlin, Python, JavaScript, TypeScript, Assembly, Make, Git
 - **IoT & Networking:** MQTT, Socket.IO, SSDP, UDP/TCP Sockets, REST APIs, POP3
 - **Web & Backend:** React, NestJS, Node.js, Vite, PostgreSQL, Prisma ORM, Redis, JSON Schema/Ajv
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PROJECTS

AUTOSAR Adaptive Front-lighting System

- Developed a real-time Adaptive Front-lighting System using SWCs, Runnables, Mode Management, and RTE interfaces (Sender-Receiver & Client-Server).
- Engineered dynamic lighting algorithms driven by steering wheel position, ambient light sensors, battery voltage, and Passing Lane configurations.
- Diagnostics & Memory (BSW): Integrated DCM, DEM, and NvM stacks for UDS diagnostic handling (DIDs, DTCs, ignition cycles) and thread-safe persistent storage using PIM and RTE Exclusive Areas.
- Testing & Tools: Validated ECU software behavior using AUTOSAR configuration tools, CANoe bus simulation, and hardware debugging tools under simulated driving conditions.

Linux Kernel Driver

- Developed a Linux platform and misc kernel driver for Raspberry Pi 2, using Device Tree configuration, GPIO, hardware timer interrupts, and a blocking write-interface to control LED signal sequences.
- Implemented runtime configuration through IOCTL, interrupt-driven buffer processing, and spinlock-based synchronization to protect shared resources and prevent race conditions/deadlocks.

MAVN-to-MIPS Compiler

- Implemented a C++ compiler that translates a custom MAVN language into MIPS assembly, using lexical analysis and a recursive-descent parser.
- Developed liveness analysis, interference graph construction, and register allocation using graph coloring, and added custom SEQ, BGE, and DIV instructions.

Data Center Algorithms & Network Routing Python

- Implemented Python data structures and algorithms (BST, Insertion sort, Dijkstra) with performance testing and visualization using Matplotlib.

FX Audio Processing Module

- Developed and integrated a C/C++ multichannel DSP audio FX module with configurable IIR high-pass filtering, gain, delay, mute, and upmix, validated using Wave Compare, GRANT and CUTE testing tools.
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ADDITIONAL

- **Languages:** English (fluent), German (B1), Serbian (native)
- Worked in 3 summers in retail, handling cash and inventory, developing strong customer service skills and learning to be a reliable team player.
- **Interests:** Hiking, Running, Reading, Chess